Software Requirements Specification

for

Online Traffic Fine Payment System

(OTFPS – JURMANA)

Version 1.0 approved

Prepared by Ragyam Sinha

TMSL-CSE

02.08.2025

Table of Contents

Table of Contents

1. Introduction 4
   1.1. Purpose/Objective 4
   1.2. Document Conventions (Definition, Acronyms, Abbreviations) 4
      1.2.1. OTP: One Time Password 4
      1.2.2. EV: Electric Vehicle 4
      1.2.3. UI: User Interface 4
      1.2.4. Admin: Traffic Police or Authorized Government Official 4
      1.2.5. User: Citizen or vehicle owner 4
      1.2.6. System: Online Traffic Fine Payment Web Application 4
   1.3. Scope 4
   1.4. References 4
2. History/Background Study (Sources of Domain Knowledge) 4
   2.1. Technical Literature 4
   2.2. Existing Applications 4
   2.3. Customer Surveys 6
   2.4. Expert Advice 6
   2.5. Current/Future requirements 9
3. Overall Description 9
   3.1. Product Functions 9
      3.1.1. Hardware Requirement 9
      3.1.2. Client Side: Any modern device with internet (PC/Laptop/Mobile) 9
      3.1.3. Software Requirement 9
      3.1.4. Client Side: Web browser (Chrome, Edge, Firefox), OS: Windows / Android / iOS / Linux 9
   3.2. Functional Requirements 10
      3.2.1. User Registration and Authentication 10
      3.2.2. OTP Verification 10
      3.2.3. User Login 10
      3.2.4. Admin Login 10
      3.2.5. Fine Management 10
      3.2.6. Online Payment 11

3.2.7. Violation Management 11
3.2.8. Notification System 12
3.3. Non-Functional Requirements 13
3.3.1. Correctness Requirement 13
3.3.2. Portability requirement 13
3.3.3. Efficiency Requirement 14
3.3.4. Usability Requirement 14
3.3.5. Reusability Requirement 14
3.3.6. Reliability Requirement 14
3.3.7. Maintainability Requirement 14
3.4. User Characteristics 14
3.5. Design & Implementation Constraints 14
3.6. Assumptions & Dependencies 15
4. Interface Requirements 15
4.1. User Interfaces 15
4.2. Hardware Interfaces 15
4.3. Software Interfaces 15
4.4. Communication Interfaces 15
5. Conclusion 16

1. Introduction

1.1. Purpose/Objective

The purpose of this document is to define the software requirements for the development of an **Online Traffic Fine Payment System**. The system will allow citizens to view and pay traffic violation fines online, enable authorities to issue and manage fines digitally, and streamline the overall traffic enforcement process.

1.2. Document Conventions (Definition, Acronyms, Abbreviations)

- OTP: One Time Password
- EV: Electric Vehicle
- UI: User Interface
- Admin: Traffic Police or Authorized Government Official
- User: Citizen or vehicle owner
- System: Online Traffic Fine Payment Web Application

1.3. Scope

This project provides an online platform for traffic fine management in urban and rural areas. It supports user registration, fine notification, online payments, violation tracking, and dispute handling. The system ensures transparency, reduces manual errors, and integrates with national vehicle databases and e-wallets.

It includes features for both **citizens** (view/pay/dispute fines) and **admins** (issue fines, review disputes, generate reports). Special rules for **Electric Vehicles** are also incorporated.

1.4. References

Indian Motor Vehicle Act, 1988:- https://morth.nic.in/en/motor-vehicles-act-1988
Government e-Challan portal:- https://echallan.parivahan.gov.in/
Vahan: - https://vahan.parivahan.gov.in/nrservices/

2. History/Background Study (Sources of Domain Knowledge)

2.1. Technical Literature

Research papers and government guidelines on traffic enforcement automation and smart city integration were studied to design system modules and ensure legal compliance.

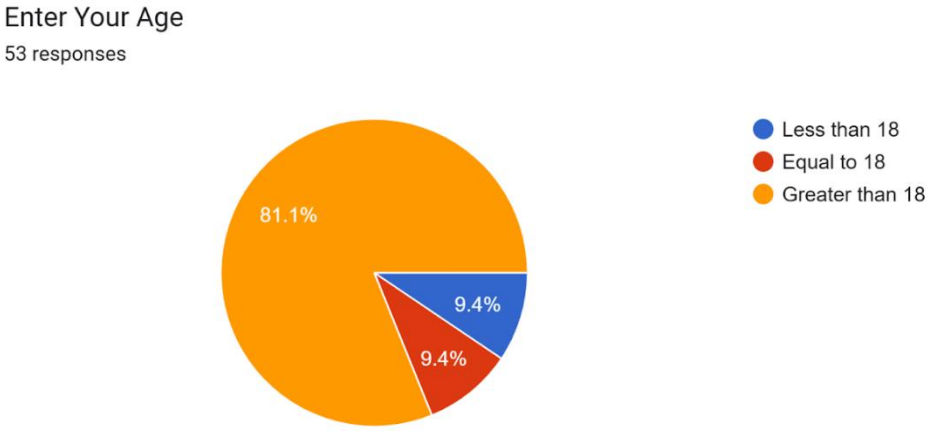
2.2. Existing Applications

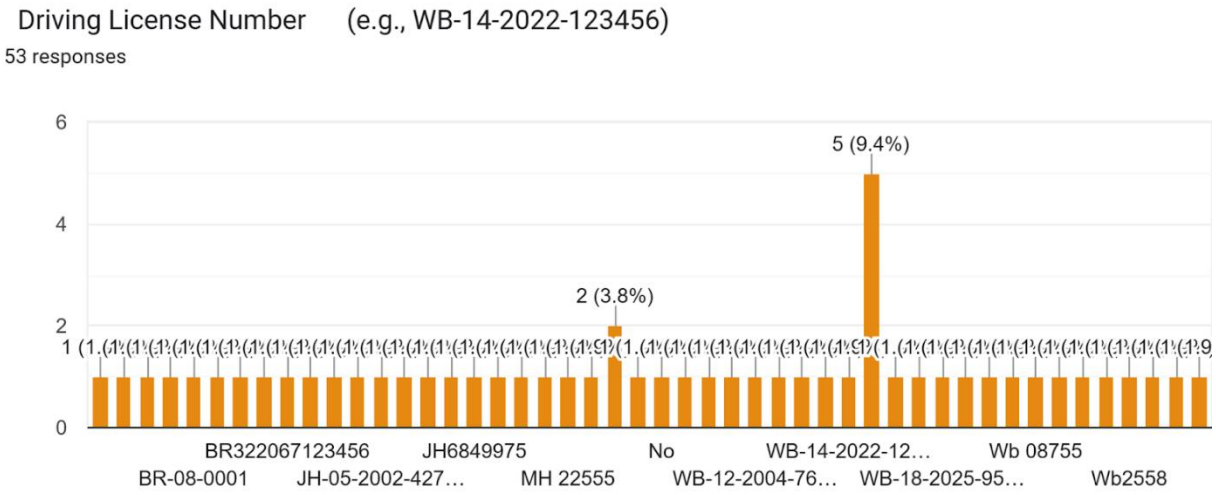
Category	Traditional Fine System	Online Fine Payment System
Mode of Payment	Cash only at police booths or courts	UPI, Net Banking, Credit/Debit Card, Wallets
User Convenience	Requires physical presence, time-consuming	24/7 access from anywhere
Registration	Manual documentation	Digital registration with OTP verification
Fine Issuance	Paper challan with handwritten details	Auto-generated digital challan with media proof
Notifications	No real-time alerts	SMS/Email notifications for fines and updates
Payment Confirmation	Manual receipt	Instant digital receipt + history in dashboard
Dispute Handling	Long queues, physical application	Online dispute submission and tracking
Admin Efficiency	Manual record keeping, prone to loss	Centralized dashboard, filters, and reports
Security	Risk of human error or tampering	OTPs, encrypted transactions, role-based access
Tracking & Reports	Paper logs and manual reports	Instant reports (paid/unpaid/disputed fines)
Support for EV Rules	Not available or inconsistent	Automated waiver checks for EVs
Environmental Impact	Uses paper for challans and receipts	Paperless system (eco-friendly)
Fraud Prevention	Higher risk (cash mishandling, fake slips)	Audit trails, secure payments, verified IDs
Cost of Implementation	Low tech cost, high manpower	High initial tech cost, low operational manpower
Public Transparency	Limited user insight	Transparent fine records, proof-based violations

We analyzed government portals like *e-Challan*, *Delhi Traffic Police Portal*, and apps like *paravian* to understand their strengths and limitations.

2.3. Customer Surveys

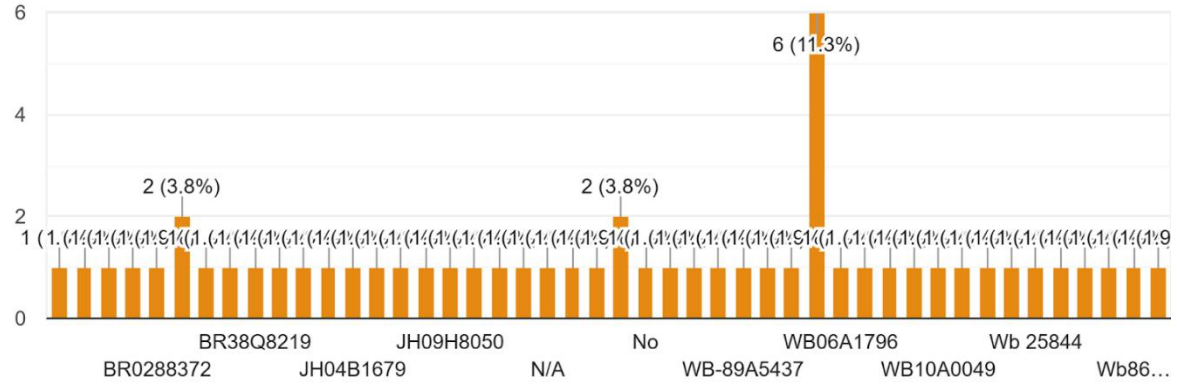
- Click here to view Customer form.
- Click here to view Response Sheet.





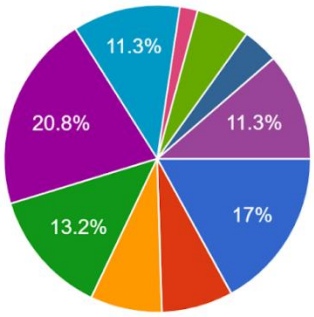
Vehicle Registration Number (e.g., WB06A1234)

53 responses



Select Violation Type(s)

53 responses

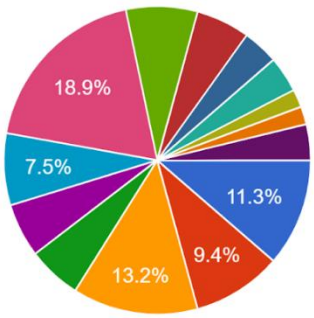


- Over-speeding
- Signal jumping
- Wrong parking
- No seatbelt
- No helmet
- Using phone while driving
- Driving without license
- Pollution Under Control (PUC) expired

1/2

Fine Amount As per violation

53 responses

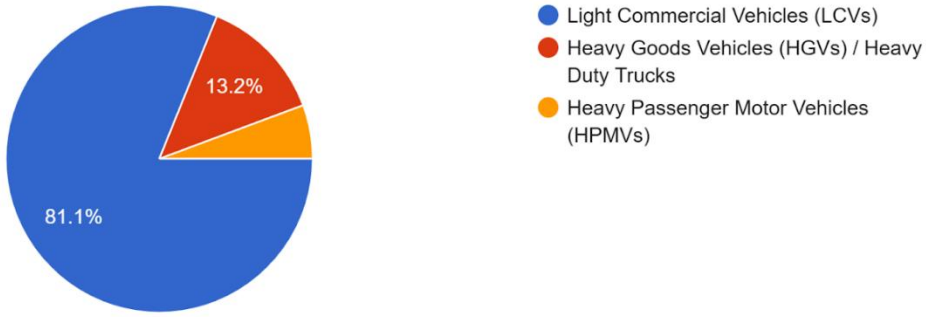


- Over-speeding – ₹1,000 for LMV, ₹…
- Signal Jumping – ₹1,000
- Wrong Parking – ₹500 (general ar…
- Driving without valid Driving Licen…
- Using Mobile Phone while Driving…
- Driving Against One-Way / Wrong…
- Not Wearing Helmet (2W) / Seatb…
- Pollution Certificate (PUC) Expired…

1/3

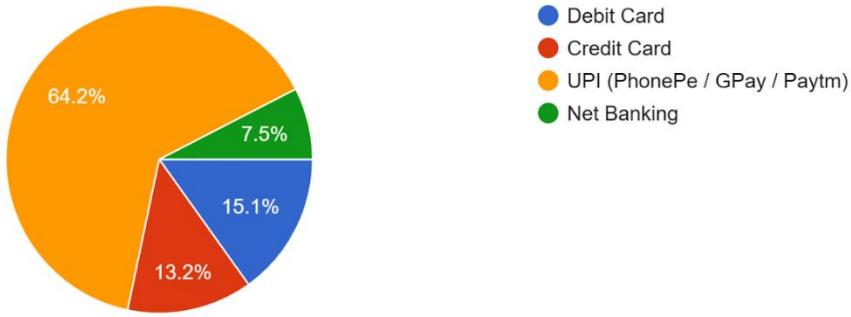
Vehicle Type's

53 responses



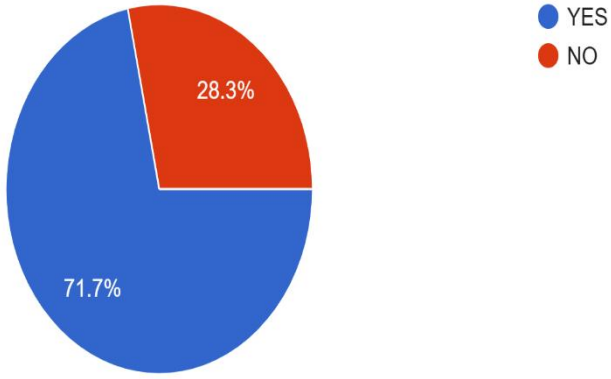
Payment method

53 responses



Do you want a digital receipt?

53 responses



2.4. Expert Advice

Consultation with traffic police officers and municipal IT professionals helped us identify the practical workflow for issuing fines and handling disputes.

2.5. Current/Future requirements

Current requirements include online payment, user dashboard, admin……panel, and dispute…….module. Future additions may include AI-based violation detection from CCTV footage and mobile app integration.

3. Overall Description

3.1. Product Functions

3.1.1.1. Hardware Requirement

- ❖ Client Side: Any modern device with internet (PC/Laptop/Mobile)

- Server Side:
  i. Processor: Intel Xeon or AMD equivalent
  ii. RAM: Minimum 8 GB
  iii. Storage: 200 GB SSD
  iv. Internet: 100 Mbps bandwidth
  v. Backup Power and Cooling

3.1.2. Software Requirement

- ❖ Client Side: Web browser (Chrome, Edge, Firefox), OS: Windows / Android / iOS / Linux

- Server Side:
  i. OS: Ubuntu Server 20.04 LTS
  ii. Web Server: Apache / Nginx
  iii. Backend: Node.js / Java / Python (Django)
  iv. Database: MySQL / PostgreSQL
  v. APIs: Government vehicle DB API, Payment Gateway API
  vi. Security: SSL/TLS, CAPTCHA, JWT Authentication

3.2. Functional Requirements

3.2.1. User Registration and Authentication

Description: Allows a new user to register on the system with their personal and vehicle details.
Input: Name, Mobile Number, Email ID, Vehicle Registration Number, Password.
Output: Account created message or validation error; OTP sent for verification.
Error: mobile/email exists, invalid input, vehicle number wrong, fields empty

3.2.2. OTP Verification

Description: Verifies the user account through OTP sent to registered mobile/email.
Input: OTP (One-Time Password).
Output: Confirmation of successful verification or failure message.
Error: wrong otp, otp expired, too many attempts.

3.2.3. User Login

Description: Allows registered users to log in securely to their account.
Input: Email/Phone Number, Password.
Output: Successful login with dashboard access or error message for invalid credentials.
Error: wrong credentials, not verified, login limit reached

3.2.4. Admin Login

Description: Allows traffic officers or authorized personnel to log in as an admin.
Input: Admin Username and Password.
Output: Access to the admin dashboard or error message.
Error: invalid login, not authorized, account blocked.

3.2.5. Fine Management

3.2.5.1. Fine Issuance

Description: Enables traffic authorities to issue new fines for traffic violations.

Input: Vehicle Number, Violation Type, Date & Time, Location, Fine Amount, Proof (image/video).
Output: Fine record generated and stored; user notified.
Error: vehicle not found, no proof, missing info, duplicate fine.

3.2.5.2. Fine Tracking

Description: Allows users to view all fines related to their vehicle, along with payment and dispute statuses.
Input: User ID or Vehicle Number.
Output: List of fines with status (Paid/Unpaid/Disputed), fine details.
Error: no fines found, wrong vehicle number

3.2.6. Online Payment

Description: Enables users to pay their traffic fines securely via online payment gateways.
Input: Fine ID, User ID, Payment Method (UPI/Card/Net-banking), Payment Details.
Output: Payment confirmation, receipt generated, and fine status updated to "Paid".
Error: payment failed, invalid details, already paid, timeout.

3.2.7. Violation Management

3.2.7.1. View Violation Details

Description: Displays detailed information of the violation for which the fine was issued.
Input: Fine ID or Vehicle Number.
Output: Violation type, date/time, location, photographic evidence, amount.
Error: fine not found, invalid input.

3.2.7.2. Dispute Submission

Description: Allows users to submit a complaint if they believe the fine was wrongly issued.
Input: Fine ID, User ID, Reason for dispute, Optional supporting document/image.
Output: Acknowledgment of dispute and tracking ID.
Error: fine not found, already disputed, reason missing.

3.2.7.3. Dispute Resolution

Description: Enables admins to review disputes and provide resolutions.

Input: Dispute ID, Admin Comments, Status Update (Accepted/Rejected/Under Review).
Output: Dispute status updated; user notified.
Error: invalid dispute id, update failed, comment missing.

3.2.8. Notification System

3.2.8.1. Fine Notification

Description: Sends automated SMS/Email notifications to the user when a fine is issued.
Input: Fine ID, User Contact Details.
Output: Notification message sent with fine summary and payment link.
Error: contact missing, message not sent

3.2.8.2. Payment Confirmation Notification

Description: Sends confirmation message after successful payment of fine.
Input: Payment ID, Fine ID, User ID.
Output: Digital receipt sent via SMS/Email.
Error: payment not found, failed to notify

3.2.8.3. Dispute Status Notification

Description: Sends alert when the status of a raised dispute is updated.
Input: Dispute ID, User ID, Status Update.
Output: Message with updated status and admin remarks sent to user.
Error: payment not found, failed to notify.

3.2.9. Dashboard & Reports

3.2.9.1. User Dashboard

Description: Provides users with a personal dashboard showing fines, payment history, and dispute records.
Input: User ID.
Output: Dashboard view with categorized fine details, total dues, and status tracking.
Error: user not found, session expired

3.2.9.2. Admin Dashboard

Description: Allows traffic officials to view and manage issued fines, disputes, payment reports, and violation trends.
Input: Admin ID.
Output: Interactive dashboard with filters for region, time period, vehicle type, and violation category.
Error: access denied, load failed.

3.2.9.3. Report Generation

Description: Allows admin to generate custom reports on fines collected, unpaid fines, disputed fines, and vehicle violation stats.
Input: Date range, filter options (location/type/status).
Output: Exportable reports in PDF/Excel format.
Error: invalid date, no data, export failed

3.2.10. Electric Vehicle (EV)-Specific Rules

3.2.10.1. EV Fine Waiver Check

Description: System checks for EV-specific exemption policies before applying a fine.
Input: Vehicle Registration Number.
Output: Decision on whether fine is waived or applicable based on policy.
Error: policy not found, wrong vehicle type

3.2.10.2. EV Priority Handling

Description: Flags EV-related fines for manual review if applicable under government incentives or special exemptions.
Input: Fine ID, Vehicle Type.
Output: Fine marked for review; notification sent to admin.
Error: invalid input, queue full.

3.2.10.3. EV Fine Summary

Description: Provides summary of fines applicable specifically to electric vehicles under current legal framework.
Input: Filter: Vehicle Type = EV.
Output: List of fines, status, and whether they were exempted or paid.
Error: no data found, filter failed.

3.3. Non-Functional Requirements

3.3.1. Correctness Requirement

The system must ensure accurate recording of user data, fines, payment transactions, and violation details. Any incorrect data entry should be rejected or flagged. Fine amounts must match the violation category defined by government rules.

3.3.2. Portability requirement

The system should be accessible on various platforms including desktops, laptops, tablets, and smartphones via modern web

browsers. It must support multiple operating systems like Windows, Android, iOS, and Linux.

3.3.3. Efficiency Requirement

The application should respond to user actions within 2 seconds and process payments in under 5 seconds. Server-side APIs must return data within acceptable timeframes even during peak hours.

3.3.4. Usability Requirement

The UI should be simple, user-friendly, and accessible to users of all age groups. Icons, labels, and instructions should be clear and use local languages where possible. The system should offer an intuitive dashboard for both users and admins.

3.3.5. Reusability Requirement

Modules like login authentication, payment gateway integration, and notification services should be designed as reusable components to support integration in future government apps or portals.

3.3.6. Reliability Requirement

The system must maintain high availability (uptime > 99.5%) and ensure data is not lost during failures. Proper backup mechanisms must be in place, and the system should recover gracefully from unexpected crashes.

3.3.7. Maintainability Requirement

The system codebase and architecture should follow modular and layered design principles, enabling future updates, bug fixes, and enhancements with minimal effort.

3.4. User Characteristics

General Users (Citizens): Non-technical users with basic internet knowledge, using the platform to view and pay fines.
Traffic Officers (Admins): Familiar with traffic rules and violation data entry. May require training for system use.
IT Support/Admins: Technically trained users responsible for system maintenance and monitoring.

3.5. Design & Implementation Constraints

Integration must follow government security standards (e.g., Digi Locker, Parivahan APIs).
Must comply with privacy laws like the IT Act, 2000, and protect sensitive user information.
Only approved government payment gateways may be used.
Development must be done using open-source technologies where possible.

3.6. Assumptions & Dependencies

Users have access to the internet and a mobile number linked to their vehicle.
Vehicle and license data will be retrieved from a government-maintained central database.
SMS/Email services from third-party providers must be reliable for OTP and alerts.
Payment gateway APIs must remain operational and up-to-date.

4. Interface Requirements

4.1. User Interfaces

Login/Signup Page: Simple form with OTP-based validation.
User Dashboard: Lists fines, payment status, and option to raise disputes.
Admin Panel: Interface to issue fines, manage disputes, and generate reports.
Responsive Design: Layout adjusts automatically for mobile and desktop screens.

4.2. Hardware Interfaces

Integration with traffic officers' devices (tablets, mobile phones) for issuing fines.
Optional: CCTV cameras or image processing tools for violation detection.
Print interface for generating physical fine receipts (optional for admin).

4.3. Software Interfaces

Government Vehicle Database (e.g., Parivahan)
Payment Gateway APIs (Paytm, Razorpay, BHIM UPI)
SMS/Email Services (Twilio, SendGrid, etc.)
NIC or Smart City Platform Integration APIs (optional)

4.4. Communication Interfaces

HTTPS Protocol for secure data transfer.
REST APIs for communication between frontend, backend, and external systems.
Email/SMS Channels for real-time alerts and OTPs.
WebSocket (optional) for admin live updates on violations.

5. Conclusion

The **Online Traffic Fine Payment System** is aimed at modernizing the traffic violation and fine payment process through digitization. It ensures transparency, ease of use, and efficient administration while integrating with existing government databases. With strong functional and non-functional requirements, this system will serve as a secure and scalable solution for urban traffic management.